

Linear-filter analysis of sparse conductance-driven pixel features

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Abstract

We independently simulated and analyzed the conductance-filtered single-pixel feature used in rate-coded MNIST experiments. A stationary linearization explains the flat low-frequency gain, the high-frequency roll-off, and the lobes introduced by finite-window averaging. It preserves the ordering and broad curvature of empirical response means but overpredicts their magnitude; its predicted feature SD also peaks too early. The discrepancy arises because 0–5 expected events form discrete spike-count and spike-time mixtures rather than small Gaussian fluctuations around a stationary operating point.

Purpose

This standalone experiment asks:

What does stationary linear-filter theory explain about the finite-window conductance-driven pixel feature, and where does that approximation fail under sparse input?

The experiment specifies and simulates its own physical model. It shares no artifacts or experiment code with EXP080; the two entries are related only by studying the same parameterized feature family.

Methods

Independent finite-window simulation

Each condition was defined directly by its expected event count $\Lambda \in [0, 5]$ during a $T = 200$ ms presentation. The per-timestep event probability was

$$p_{\text{event}} = \frac{\Lambda}{N} = \lambda \Delta \frac{t}{1000}, \quad (1)$$

where $N = \frac{T}{\Delta t}$, $\Delta t = 0.1$ ms, and $\lambda = 1000 \frac{\Lambda}{T}$ spikes/s. This direct expected-count grid avoids duplicating conditions through different rate–intensity pairs.

Conductance and membrane voltage followed

$$g(t) = \beta g(t - \Delta t) + wS(t), \quad (2)$$

$$\beta = \exp\left(-\Delta \frac{t}{\tau_{\text{AMPA}}}\right), \quad (3)$$

$$C \frac{dv}{dt} = g_L(E_L - v) + g(t)(E_e - v). \quad (4)$$

We used $C = 1$ nF, $g_L = 0.05$ μ S, $E_L = -65$ mV, $E_e = 0$ mV, $\tau_{\text{AMPA}} = 2$ ms, and independent probe conductances $w \in \{0.6, 1.2, 2.4\}$ μ S. Every presentation began from $g(0) = 0$ and $v(0) = E_L$. Its scalar feature was

$$z = \frac{1}{T} \int_0^T (v(t) - E_L) dt. \quad (5)$$

At each of 101 unique expected-count conditions and each conductance, we generated 512 new presentations. Separate Separate simulations with 4096 draws characterized representative full response distributions.

Stationary operating point

For stationary Poisson rate λ , the mean AMPA conductance is

$$\bar{g}_\lambda = \lambda w \frac{\tau_{\text{AMPA}}}{1000}. \quad (6)$$

Replacing the fluctuating conductance by this mean gives the operating-point voltage

$$\bar{v}_\lambda = \frac{g_L E_L + \bar{g}_\lambda E_e}{g_L + \bar{g}_\lambda}. \quad (7)$$

Because this operating point is constant across the window, its predicted mean feature is

$$\mu_{\text{linear}}(z) = \bar{v}_\lambda - E_L. \quad (8)$$

Local frequency response

Linearizing Equations 2–4 around the operating point gives the response from input-rate fluctuation to voltage fluctuation,

$$G_\lambda(\omega) = \frac{w}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \cdot \frac{E_e - \bar{v}_\lambda}{i\omega C + g_L + \bar{g}_\lambda}. \quad (9)$$

Here ω is angular frequency in rad/ms. Averaging voltage over the finite presentation contributes

$$A_T(\omega) = \frac{1 - \exp(-i\omega T)}{i\omega T}, \quad (10)$$

so the complete input-to-feature response is

$$H_\lambda(\omega) = A_T(\omega) G_\lambda(\omega). \quad (11)$$

We evaluated Equations 9 and 11 from 0.1–200 Hz at 0.25, 3, and 25 spikes/s for the nominal 1.2 μ S probe. Frequency in Hz was converted using $\omega = 2\pi \frac{f}{1000}$. Gain was reported relative to the low-drive DC response,

$$20 \log_{10} \left(\frac{|G_\lambda(2\pi \frac{f}{1000})|}{|G_{\lambda_{\text{low}}}(0)|} \right). \quad (12)$$

Linearized feature variance

The centred ideal Poisson input has a white two-sided spectrum on the millisecond time base,

$$S_{\text{in}}(\omega) = \frac{\lambda}{1000}. \quad (13)$$

The feature spectrum and variance are therefore

$$S_z(\omega) = |H_\lambda(\omega)|^2 S_{\text{in}}(\omega), \quad (14)$$

$$\text{Var}_{\text{linear}}(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} |H_\lambda(\omega)|^2 S_{\text{in}}(\omega) d\omega. \quad (15)$$

We integrated Equation 15 numerically on a logarithmic frequency grid. A second calculation with half as many points measured quadrature refinement.

Results

Empirical finite-window response

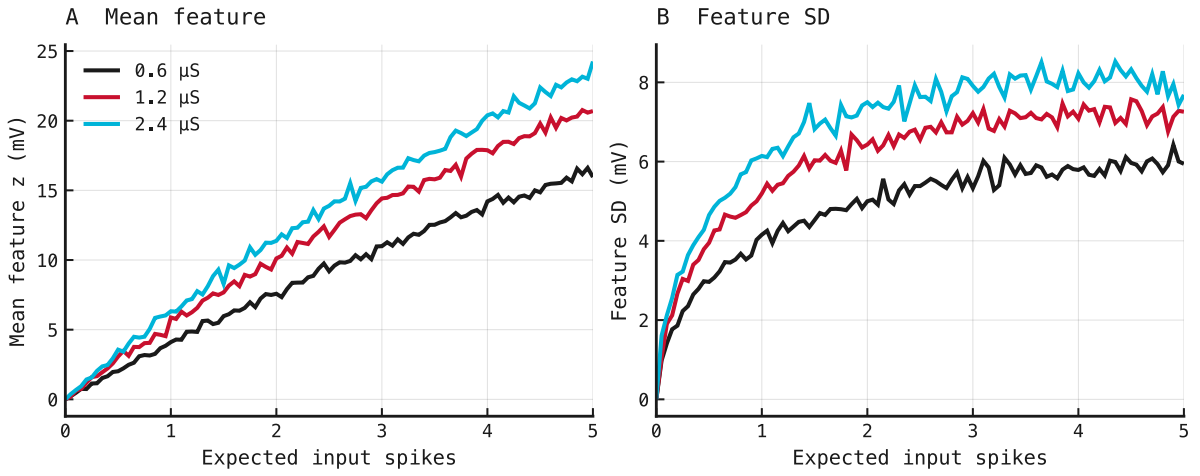


Figure 1: Empirical finite-window response over a unique expected-event-count grid. Panel A shows mean feature; Panel B shows feature SD. Every point summarizes 512 independently simulated presentations. Black, red, and cyan denote 0.6, 1.2, and 2.4 μS conductance increments.

Mean response increased with expected event count and event strength. Variability initially increased as spike count and timing diversified, then flattened or declined as relative count fluctuations, shunting, and voltage saturation became more important.

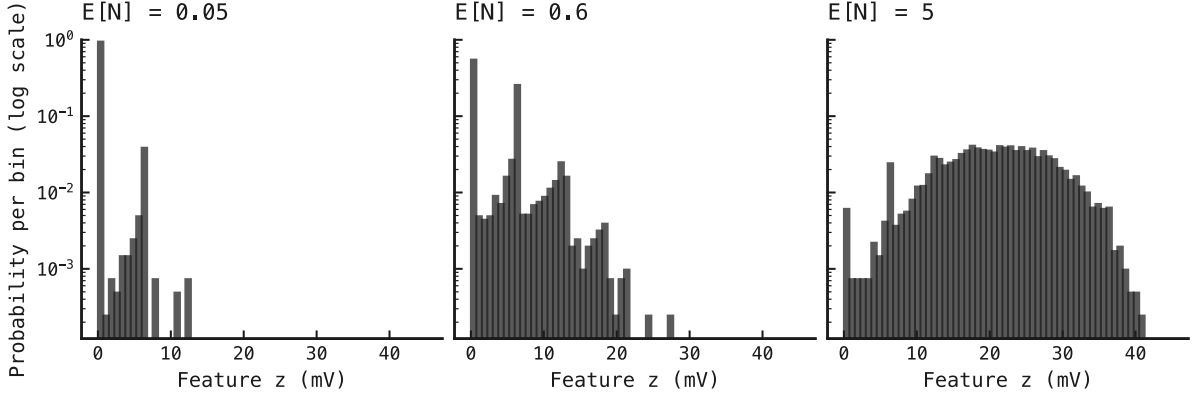


Figure 2: Empirical feature distributions at 0.05, 0.6, and 5 expected events for the nominal $1.2 \mu\text{S}$ probe. Each panel contains 4096 new presentations. Bars report probability per common fixed-width bin on a shared logarithmic axis. Sparse input produces an atom at rest and separated timing-dependent responses; increasing event count produces a smoother distribution.

An expected count is not an observed count. At $\Lambda = 1$, for example, presentations can contain zero, one, two, or more events. Even at fixed count, their event times alter the finite-window average.

Analytical frequency response

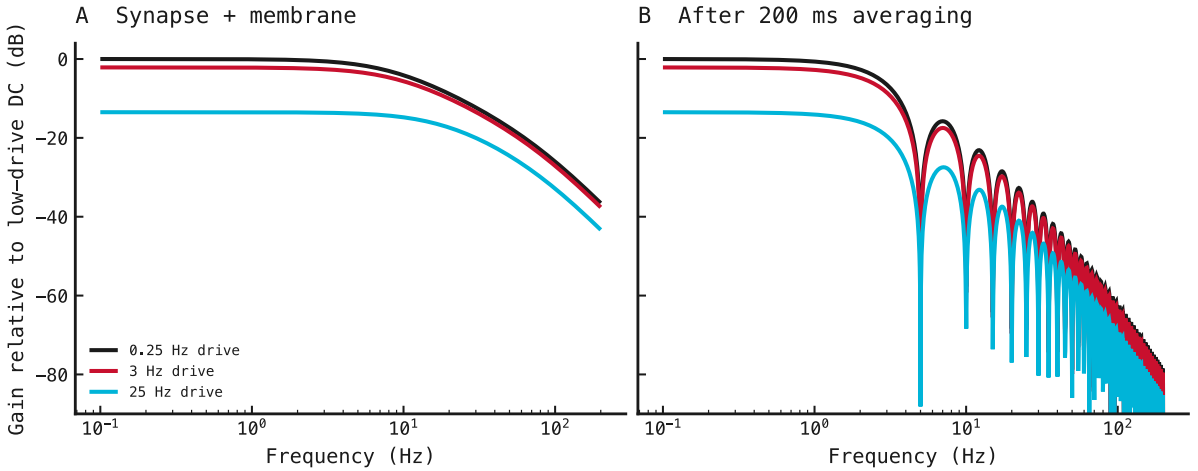


Figure 3: Analytical linearized frequency response at the nominal $1.2 \mu\text{S}$ probe. Black, red, and cyan denote operating rates 0.25, 3, and 25 spikes/s. Panel A shows Equation 9; Panel B shows the complete 200 ms window-averaged response from Equation 11.

In Panel A, each curve is flat at low frequency because the AMPA conductance and membrane voltage can follow a slowly varying input quasi-statically. Greater drive lowers that plateau by shunting the membrane and reducing the excitatory driving force. The curves bend downward once modulation becomes too fast for the synaptic and membrane timescales to follow.

Panel B is the same synapse-plus-membrane response multiplied by the rectangular-window response. With $T_s = \frac{T}{1000}$ s,

$$\left| A_T \left(2\pi \frac{f}{1000} \right) \right| = \left| \frac{\sin(\pi f T_s)}{\pi f T_s} \right|. \quad (16)$$

Its zeros at $f = \frac{n}{T_s}$ cancel complete cycles at 5, 10, 15 Hz, and so on. The absolute sinc response creates the lobes, while the Panel A low-pass response supplies their falling envelope.

Analytical and empirical moments

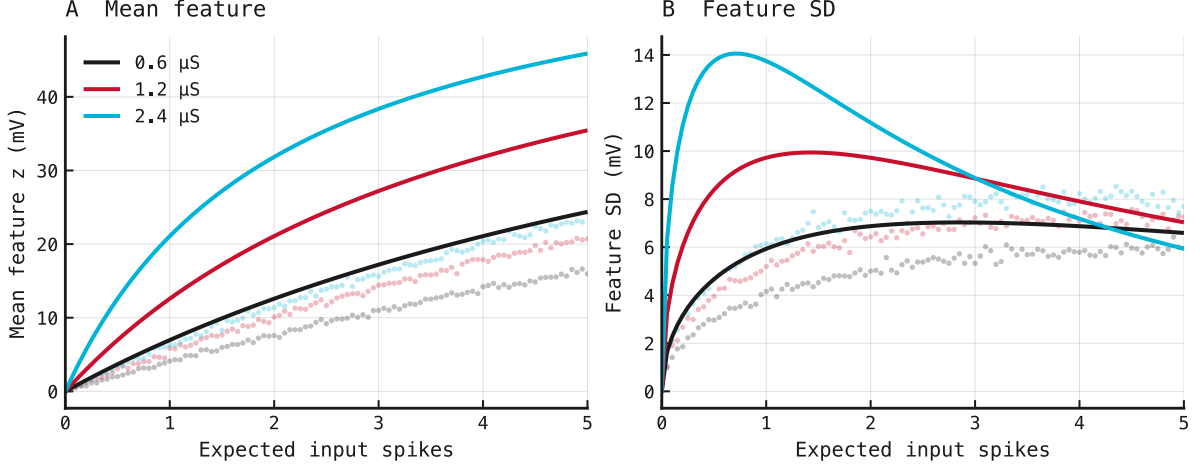


Figure 4: Analytical and empirical moments over expected input events. Solid curves are stationary predictions from Equations 8 and 15; faint points are independently simulated finite-window estimates. Panel A shows mean feature and Panel B feature SD.

The analytical mean preserved the broad curvature and conductance ordering but exceeded the empirical magnitude. Its median predicted-to-empirical ratio was 1.972. The stationary calculation evaluates the response at mean conductance, $z(E[g])$, whereas simulation estimates the response over random conductance paths, $E[z(g)]$. Saturation makes these unequal, approximately

$$E[z(g)] < z(E[g]).$$

The stationary input also acts throughout the window; a real late event contributes only briefly to Equation 5.

The SD mismatch is more fundamental. The empirical distribution is a spike-count and spike-time mixture,

$$p_Z(z) = \sum_{n=0}^{\infty} P(N = n) p_{Z|N}(z|n). \quad (17)$$

Equation 15 instead treats input as a small stationary fluctuation around one operating point. At low expected counts, zero-event trials remain exactly at rest, one large event produces a timing-dependent response, and multiple events interact through shunting and saturation. That non-Gaussian mixture explains why the analytical SD rises too sharply and peaks too early, especially for the largest conductance increment.

Conclusion

Stationary linear-filter theory correctly explains the low-frequency plateau, high-frequency roll-off, averaging-window zeros, and qualitative operating-point dependence. It does not quantitatively predict finite-window moments in the sparse, large-jump regime. Direct simulation is therefore required for empirical rate selection, while the analytical model remains useful for explaining the filter’s structure.

Limitations

The empirical moment grid used 512 presentations per condition; it was designed to resolve the qualitative theoretical comparison, not to estimate extreme distribution tails. The analytical variance assumes ideal continuous-time Poisson drive and a local stationary linearization, whereas simulation uses discrete Bernoulli events and begins from rest.

Reproducibility

`uv run python experiments/exp081.py` regenerates every sample, summary, and figure. The runner contains a complete independent physical specification and consumes no artifacts or code from another experiment.

Appendix A: derivation of the analytical filter

A.1 Stationary conductance and voltage

Write the continuous AMPA equation driven by an event train $s(t) = \sum_k \delta(t - t_k)$ as

$$\frac{dg}{dt} = -\frac{g}{\tau_{\text{AMPA}}} + ws(t). \quad (\text{A1})$$

For stationary rate λ spikes/s, the rate on the millisecond time base is $\frac{\lambda}{1000}$. Taking expectations of Equation A1 and setting the stationary derivative to zero gives

$$0 = -\frac{\bar{g}_\lambda}{\tau_{\text{AMPA}}} + w\frac{\lambda}{1000},$$

hence Equation 6. The stationary membrane equation is

$$0 = g_L(E_L - \bar{v}_\lambda) + \bar{g}_\lambda(E_e - \bar{v}_\lambda), \quad (\text{A2})$$

and collecting the terms multiplying $\bar{v}_\lambda$ gives Equation 7.

A.2 Synapse-plus-membrane linearization

Decompose each quantity into its operating point and a small fluctuation,

$$g = \bar{g}_\lambda + \delta g, \quad v = \bar{v}_\lambda + \delta v, \quad s = \frac{\lambda}{1000} + \delta s.$$

Substitution into Equation A1 and cancellation of the stationary terms gives

$$\frac{d\delta g}{dt} = -\delta \frac{g}{\tau_{\text{AMPA}}} + w\delta s. \quad (\text{A3})$$

With Fourier convention $\frac{d}{dt} \rightarrow i\omega$,

$$\delta g(\omega) = \frac{w}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \delta s(\omega). \quad (\text{A4})$$

Substitute the decompositions into Equation 4. After cancelling Equation A2 and discarding the second-order product $\delta g \delta v$,

$$C \frac{d\delta v}{dt} = -(g_L + \bar{g}_\lambda) \delta v + (E_e - \bar{v}_\lambda) \delta g. \quad (\text{A5})$$

Fourier transformation and substitution of Equation A4 yields Equation 9. Its two denominator factors are the AMPA and membrane low-pass terms; the numerator contains the conductance increment and local excitatory driving force.

A.3 Finite-window average

Linearizing Equation 5 leaves the average of the voltage fluctuation,

$$\delta z = \frac{1}{T} \int_0^T \delta v(t) dt. \quad (\text{A6})$$

The averaging kernel is $a_T(t) = \frac{1}{T}$ for $0 \leq t \leq T$ and zero otherwise. Its Fourier transform is

$$A_T(\omega) = \frac{1}{T} \int_0^T \exp(-i\omega t) dt = \frac{1 - \exp(-i\omega T)}{i\omega T},$$

proving Equation 10. Because averaging follows the membrane response, the transfer functions multiply, proving Equation 11. Using $1 - \exp(-ix) = 2i \exp(-i\frac{x}{2}) \sin(\frac{x}{2})$ gives

$$|A_T(\omega)| = \left| \frac{\sin(\omega \frac{T}{2})}{\omega \frac{T}{2}} \right|,$$

which becomes Equation 16 after substituting $\omega = 2\pi \frac{f}{1000}$ and $T_s = \frac{T}{1000}$.

A.4 Poisson spectrum and feature variance

For ideal Poisson events with rate $\frac{\lambda}{1000}$ per ms, disjoint intervals have independent counts and count variance equals count mean. The centred impulse train therefore has autocovariance

$$R_{\delta s}(\tau) = \left(\frac{\lambda}{1000} \right) \delta(\tau). \quad (\text{A7})$$

Fourier transformation of the delta function gives the constant spectrum in Equation 13. A linear time-invariant filter multiplies the input spectrum by squared transfer magnitude, proving Equation 14. Finally, inverse Fourier transformation at zero lag gives

$$\text{Var}(z) = R_z(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_z(\omega) d\omega,$$

which, after substituting Equation 14, proves Equation 15. The result is exact for the stated stationary linearized model but only approximate for the nonlinear, start-from-rest, finite-event simulation.

References

1. Wolff & Lindner: *Mean, Variance, and Autocorrelation of Subthreshold Potential Fluctuations Driven by Filtered Conductance Shot Noise*. Neural Computation, 2010. doi:10.1162/neco.2009.02-09-958

2. Brigham & Destexhe: *Nonstationary Filtered Shot-Noise Processes and Applications to Neuronal Membranes*. Physical Review E, 2015. doi:10.1103/PhysRevE.91.062102